Upper high school · analysis and algorithms

Bracketing a root by bisection

Progressive exercises

Given information

For exercise 1: seek a zero of $f(x) = x^2 - 2$ on [1,2]. After two halvings, the retained interval is [1.25,1.5].

Exercise 1

Carry out a third step of the example.

Exercise 2

Starting with width 1, how many halvings guarantee width at most $\frac{1}{1000}$?

Exercise 3

Why does the sign change of $f(x) = \frac{1}{x}$ between -1 and 1 not prove the existence of a zero?

Exercise 4

A root lies in [1.4,1.5]. What maximum error is guaranteed by midpoint 1.45?